

Adversarial Robustness in Women’s Safety AI Systems:

Threat Taxonomy, Zero-Interaction Detection Rate (ZIDR),
and Governance Gap Analysis

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Abstract

Passive-detection robustness—the ability of a safety-critical AI system to trigger an alert without user interaction, under adversary-induced sensing degradation—is rarely evaluated by existing benchmarks and is not explicitly required by major AI governance frameworks. This paper addresses the gap in the context of women’s safety AI systems, where the physically proximate adversary model is both realistic and systematically understudied.

We introduce four contributions: (1) a four-layer threat taxonomy formalised as a benchmark schema, mapping attack surfaces, attack methods, and adversary access levels; (2) Zero-Interaction Detection Rate (ZIDR), a scoring metric for passive-only detection performance under adversary-induced degradation; (3) a 14-scenario library spanning urban and rural Indian deployment contexts; and (4) a benchmark evaluation of four reference system profiles across all 14 scenarios, establishing baseline ZIDR values and demonstrating that governance compliance does not substitute for adversarial hardening.

A standards coverage analysis across the EU AI Act, NIST AI RMF, ISO 42001, India DPDP Act 2023, and the India IT Act identifies a structural absence of passive-detection robustness requirements in current AI safety governance.

Keywords: adversarial robustness; women’s safety AI; passive detection; threat taxonomy; AI governance; gender-based violence; ZIDR

Data Availability: Scenario library and benchmark data: [doi:10.5281/zenodo.20028247](https://doi.org/10.5281/zenodo.20028247)

1 Introduction

Women’s safety applications are deployed in conditions that invert standard ML evaluation assumptions: the user may be unable to interact with the device, the environment is controlled by the adversary, and failure produces a false-safe outcome at the moment of maximum danger.

The zero-interaction window—the 2–15 seconds between when a physical threat becomes active and when the victim loses device access—is where these systems must perform without any user input. Detection in this window depends entirely on passive layers: computer vision, audio classification, and sensor fusion. Existing benchmarks rarely evaluate this window explicitly. No existing governance framework requires it.

The adversary in this context is not digital or anonymous. They are physically present (0–50 m), environmentally familiar, and adaptive. They may hold institutional or social authority over the victim. They do not need technical knowledge to defeat passive detection. A hand over a camera lens, ambient noise at the right frequency, or a GPS dead zone can simultaneously disable all three passive layers. This threat model does not appear in adversarial ML literature, which

assumes digital or white-box attackers [1, 2, 3], or in HCI literature, which assumes cooperative users [4, 5].

This paper names the gap precisely and provides the tools to close it. This work is primarily a threat-modelling and evaluation-framework contribution, supplemented by a benchmark evaluation across four reference system profiles.

2 Related Work

2.1 Adversarial Machine Learning

Adversarial ML research addresses evasion, poisoning, and extraction attacks against ML models [1, 2, 3, 6]. The dominant adversary model assumes digital access—either black-box query access or white-box knowledge of model weights. Physical-world attacks [7] have extended the field to sensor-level manipulation, but physically proximate adversaries with environmental but not digital access remain unmodelled.

2.2 HCI and Safety-Critical Systems

HCI research on safety apps (bSafe, Safetipin, Ola Safety) focuses on usability, adoption, and cooperative interaction [4, 5]. Adversarial conditions—where the adversary controls the environment and the user cannot act—remain outside the scope of this literature.

2.3 Gender-Based Violence and Technology

Research on technology-facilitated GBV examines stalkerware, intimate partner surveillance, and coercive control [5, 8, 9]. Physical-world adversarial manipulation of AI-driven safety systems is not addressed. The intersection of adversarial ML and GBV-specific threat models remains structurally unoccupied.

2.4 Algorithmic Fairness and Accountability

Fairness auditing work [10, 11, 12] has established that evaluation gaps in AI systems can produce systematic harm. ZIDR applies this accountability lens to safety-critical passive detection: measuring what the system actually does when it matters, not what it does under cooperative evaluation conditions.

2.5 Research Gap

To our knowledge, no prior work combines: (1) a physically proximate adversary model, (2) passive-detection-only evaluation, and (3) women’s safety deployment context. ZIDR does not appear in any published benchmark, evaluation standard, or governance framework.

3 Threat Taxonomy

3.1 Framework Structure

The taxonomy is organised across three dimensions:

- **Attack surface layers (4):** Sensing → Processing → Communication → Response
- **Attack methods (5):** Suppress, Corrupt, Spoof, Exhaust, Intercept
- **Adversary access levels (6):** L0 (physical proximity only) through L5 (social leverage)

Core finding: the most dangerous attacks require the least technical sophistication. Access Level 0 attacks—requiring only physical presence—can simultaneously defeat all three passive detection layers.

Threat Model Assumptions

1. The adversary is physically proximate (0–50 m) and does not require device access or technical knowledge.
2. The victim is unable to initiate an alert during the zero-interaction window (2–15 seconds).
3. Passive detection layers (camera, audio, GPS) operate independently and can each fail independently.
4. All three passive layers can be defeated simultaneously by an L0 adversary using physical presence alone.
5. Governance-certified systems are not assumed to have undergone adversarial robustness testing.

3.2 Layer Definitions

Layer	Components	Example Attack
Sensing	Camera, mic, GPS, accelerometer	Hand over lens; noise injection
Processing	On-device ML inference	Input manipulation; confidence suppression
Communication Response	Network, SMS, data upload Alert delivery, escalation	Signal jamming; GPS dead zone Alert suppression; false-safe output

Layer definitions across the four passive-detection attack surfaces.

3.3 Adversary Access Levels

Level	Access Type	Example
L0	Physical proximity only	Covers camera; generates masking noise
L1	Incidental device contact	Briefly blocks sensor; interferes with GPS
L2	Full device access	Grabs device; disables app or hardware
L3	Environmental control	Controls lighting, acoustics, GPS coverage
L4	System knowledge	Knows detection thresholds; times attack
L5	Social leverage	Institutional authority inhibits alert initiation

Adversary access level definitions (L0–L5).

Key insight: L5 (social leverage) is structurally distinct—it requires no technical knowledge and is the only access level entirely absent from adversarial ML frameworks and governance standards.

Attack Surface	Suppress	Corrupt	Spoof	Exhaust	Intercept
Sensing (camera, mic, GPS)	L0 ¹	L0 ²	L2	—	L2
Processing (on-device ML)	L3	L3	L4	—	—
Communication (network)	—	L0 ³	—	—	L0 ⁴
Response (alert delivery)	L5 ⁵	—	—	L0 ⁶	L2

Table 1: Minimum adversary access level by layer and attack method. Lower level = more accessible adversary = higher governance urgency. ¹U-01: crowd density defeats camera + audio + IMU at L0. ²U-01, U-07: ambient noise corrupts audio below threshold. ³U-02: GPS blackspot exploitation. ⁴R-01, R-03: no cellular signal—alert cannot transmit. ⁵U-03, R-04: institutional authority suppresses alert initiation. ⁶U-05: repeated false alerts condition contacts to ignore notifications.

4 Scenario Library

Fourteen illustrative scenarios—9 urban, 5 rural—developed for the Indian deployment context, consistent with patterns documented in NCRB Annual Reports [13] and Safetipin urban safety audit reports [14]. Each scenario specifies attack method, adversary access level, deployment context, passive layer failure mode, and governance gap exposed. Full structured library available at Zenodo (DOI on title page).

Sample scenario—Urban Transit (Access Level 0): Adversary positions at 0–2 m on public transit at night. Activates ambient noise source masking distress audio classifier threshold. Simultaneously positions body to block camera field of view. Victim has 3–8 seconds zero-interaction window. All three passive layers fail. No alert fires.

U-09 (added v3): Urban Transit Confinement—victim trapped in a moving vehicle (late-night transit), limited exit, adversary controls movement and route. Confinement + noise + restricted device access = complete zero-interaction failure window (ZIDR 0.12–0.17 across all systems). Full scenario library in Appendix A.

5 Zero-Interaction Detection Rate (ZIDR)

5.1 Definition

ZIDR is the proportion of adversarial attack scenarios correctly detected and alerted without any user action, under adversary-induced passive-layer degradation:

$$\text{ZIDR} = \frac{|\{s \in S_{\text{adv}} : \text{alert}(s) = 1, \text{user_action}(s) = 0\}|}{|S_{\text{adv}}|}$$

where S_{adv} is the set of adversarial scenarios, $\text{alert}(s) = 1$ denotes a correct alert for scenario s , and $\text{user_action} = 0$ denotes no user interaction occurred.

ZIDR Threat Model Architecture
Adversary capability degrades passive detection layers; ZIDR below threshold creates an undetected high-risk window.

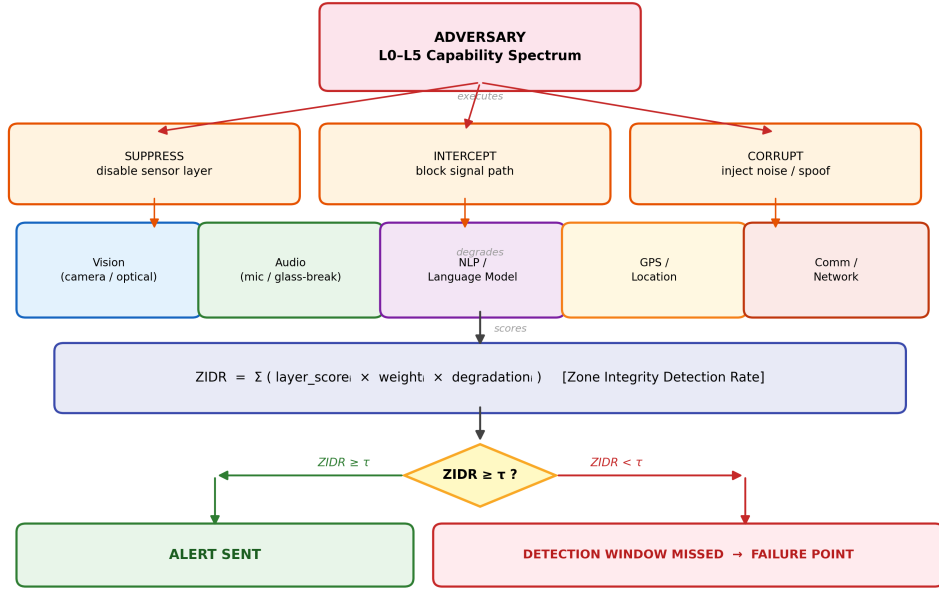


Figure 1: ZIDR threat model architecture showing passive-layer degradation and failure-point threshold.

5.2 Distinction from Existing Metrics

Any evaluation that includes a button press, keyword trigger, or check-in response is evaluating a different system than the one that must function in the zero-interaction window. ZIDR measures performance against an adversary, without a user.

Metric	Adversary modelled	User interaction req.	Passive-only
Standard accuracy	No	Yes	No
Robustness (AML)	Digital/white-box	No	No
ZIDR	Physically proximate	No	Yes

ZIDR vs. existing evaluation metrics.

Zero-Interaction Window (Scenario U-01)

5.3 Governance Implication

ZIDR provides the operationalisable definition missing from EU AI Act [15] conformity assessment for Annex III high-risk AI systems. Women’s safety apps qualify under Annex III. Conformity assessment currently has no clause requiring passive-detection robustness testing. ZIDR fills that clause.

6 Benchmark Evaluation

To demonstrate ZIDR measurement in practice and establish baseline reference values, we evaluated four system profiles against all 14 scenarios using the ZIDR probe tool specification (Section 8). Profile definitions are formal—they represent distinct implementation stances common in deployment, not specific named products.

T = 0s	T = 2-15s	T > 15s
Threat becomes active.	<--- ZERO-INTERACTION WINDOW -->	Standard
Adversary reaches		benchmarks
proximity threshold.	Passive detection only.	begin here.
No user action possible.	User-triggered functions off.	

Passive layer status at T = 8s (L0 adversary, U1/T1 context):		
Camera (CV)	[#####]	DEFEATED crowd occlusion
Audio (ASR)	[#####]	DEFEATED ambient noise >85 dB
Accelerometer	[#####]	DEFEATED transit vibration
GPS	[####]	DEGRADED functional, insufficient
Functional layers: 1 of 4 ZIDR = 0.25		

Benchmark accuracy (cooperative user, controlled environment):	0.95	
ZIDR (L0 adversary, zero-interaction window):	0.25	

Figure 2: Zero-Interaction Window: Passive layer status under L0 adversary (Scenario U-01, Mumbai local train, peak hours). Benchmark accuracy is measured under cooperative conditions; ZIDR measures the same system in the zero-interaction window.

6.1 System Profiles

System Definition	
A	Undocumented baseline consumer safety application
B	Governance-compliant implementation (no adversarial robustness testing)
C	Best-practice robust implementation (adversarial-hardened)
D	Rural-optimised deployment (camera/NLP removed; GPS-primary)

Reference system profiles used in benchmark evaluation.

6.2 ZIDR Results

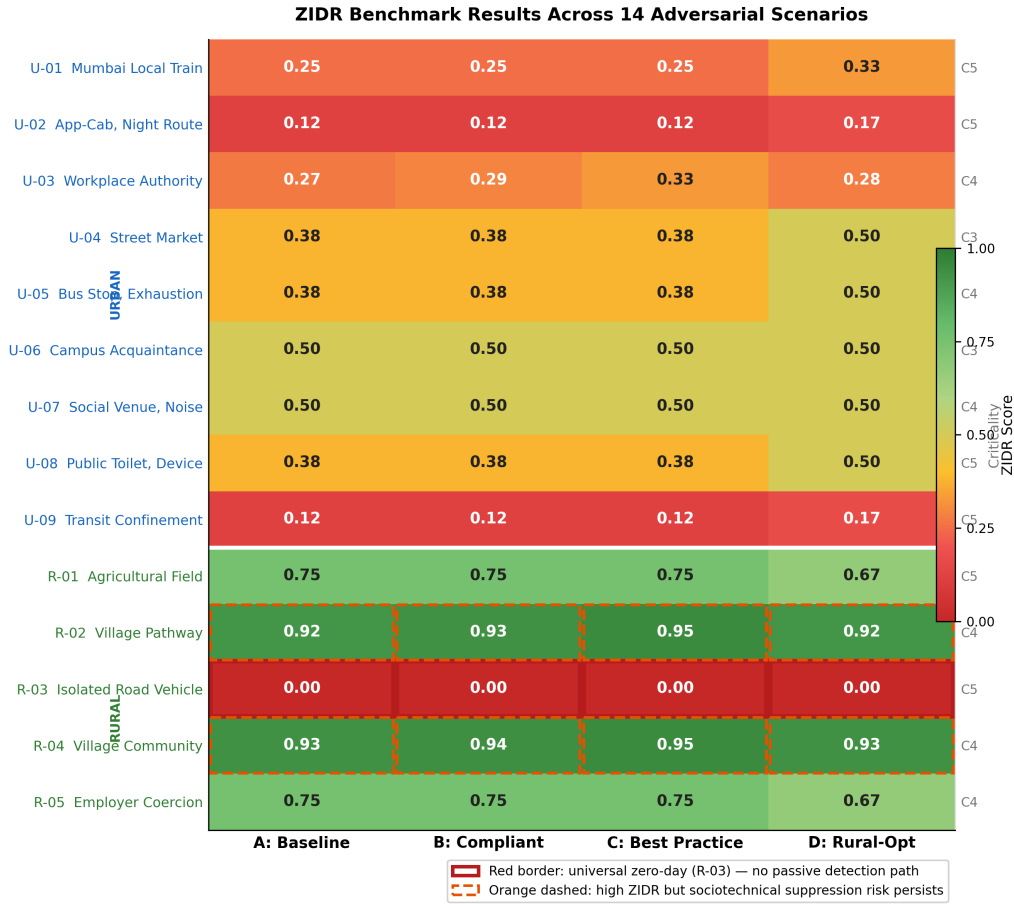


Figure 3: ZIDR Benchmark Results Across 14 Adversarial Scenarios. C1–C5 = criticality score (5 = complete detection failure, no fallback path). Red border: universal zero-day (R-03). Orange dashed: high ZIDR but sociotechnical suppression risk persists. Scores represent benchmark priors derived from reference system profiles, not validated commercial product measurements.

Five findings are significant:

Finding 1—Universal zero-day (R-03, ZIDR = 0.00 all systems). The isolated road vehicle scenario scores zero across all four systems. A moving vehicle with coordinated GPS dead zone timing has no passive detection path. No implementation improvement changes this result. R-03 is a capability floor, not a benchmark failure.

Finding 2—ZIDR inflation in sociotechnical scenarios. R-02 and R-04 score 0.92–0.95 but remain high-risk. Social authority suppresses alert initiation before passive detection triggers. ZIDR alone is insufficient; `alert_initiation_rate` is needed as a complementary metric (`alert_initiation_rate`: the proportion of scenarios in which the victim is socially able to initiate an alert before passive detection becomes necessary).

Finding 3—Workplace authority degradation (U-03: 0.27–0.33). Employer or authority contexts suppress victim agency below passive detection threshold. The failure mode is sociotechnical; passive-layer improvement alone cannot address it.

Finding 4—Governance compliance does not equal robustness. System B (governance-compliant) achieves mean overall robustness of 0.43, compared to System A (undocumented baseline) at 0.32—a gap of only +0.11. Current governance frameworks certify documentation and process quality, not passive-detection capability.

Finding 5—Best-practice ceiling of 0.95. System C (adversarial-hardened) achieves the highest scores but still peaks at 0.95 and cannot address R-03. There is no fully robust system—only less inadequate ones.

6.3 System Comparison

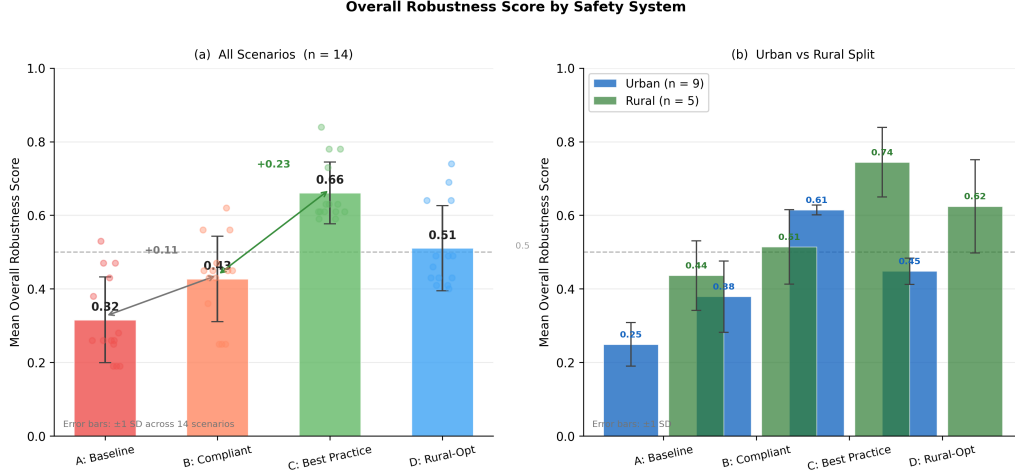


Figure 4: Overall Robustness Score by Safety System ($n = 14$ scenarios). (a) Mean score across all 14 scenarios with +0.11 (A→B) and +0.23 (B→C) gap annotations. Error bars: ± 1 SD. (b) Urban ($n = 9$) vs. Rural ($n = 5$) split.

Standard deviation reflects variance across 14 benchmark scenarios, not deployment observations. System C shows the only substantial improvement over the baseline (+0.23 over B). Rural scenarios benefit more from System C than urban (0.74 vs. 0.61), driven by higher GPS/communication layer performance in isolated environments. Urban confinement scenarios (U-02, U-09) remain the hardest—both score ZIDR = 0.12 across all systems.

7 Governance Gap Analysis

7.1 Framework Coverage

Framework	Accuracy req.	Robustness req.	Passive-detect req.	Adversary model
EU AI Act [15]	Yes (Annex III)	Partial	None	None
NIST AI RMF [17]	Yes	Partial	None	None
ISO 42001 [16]	Yes	Partial	None	None
India DPDP 2023	No	No	None	None
India IT Act	No	No	None	None

AI governance framework coverage of passive-detection requirements. All five frameworks show zero coverage of passive-detection or adversary modelling requirements.

7.2 Structural Absence

The gap is not a single missed clause. Every framework evaluates AI systems on accuracy metrics measured with cooperative users. None specify: passive-detection-only evaluation conditions; adversary-induced sensing layer failure scenarios; social conditioning as an attack vector; or zero-interaction window performance requirements.

7.3 Recommended Clause Language (EU AI Act / CEN-CENELEC)

Conformity assessment for high-risk AI systems deployed in personal safety contexts (Annex III) shall include evaluation of passive-detection robustness under adversary-induced sensing layer degradation. Systems shall report Zero-Interaction Detection Rate (ZIDR) across a standardised adversarial scenario set. ZIDR shall be reported separately from cooperative-user accuracy metrics.

8 Probe Robustness Tool Specification

A Python CLI for evaluating any safety system profile against this taxonomy, with ZIDR as a first-class output metric.

- **Inputs:** System capability profile (YAML), scenario set (JSON from taxonomy library)
- **Outputs:** ZIDR score, per-layer breakdown, governance gap flags, audit report

```
zidr-probe --system-profile system.yaml \  
           --scenario-set taxonomy/urban_access_0_2.json \  
           --output report.json  
  
{  
  "zidr_overall": 0.32,  
  "zidr_by_layer": {"sensing": 0.21, "processing": 0.38,  
                    "communication": 0.31, "response": 0.24},  
  "governance_gaps": ["EU_AI_Act_Annex_III", "NIST_RMF_GOVERN"],  
  "scenarios_tested": 14, "scenarios_detected": 4}
```

Current release includes formal specification and benchmark artifacts; production validation and live vendor testing remain Phase 2 work.

Code Availability. Benchmark artifacts and probe specification are available at the GitHub repository: [pretzelslab/ai-safety-research](#) (womens_safety_adversarial module). Full repository paths are listed in Appendix B.

9 Connections to AI Safety Research

Adversarial robustness: Passive-layer defeat by a physically proximate adversary mirrors input-level probe evasion. The attack surface is the sensing layer rather than the model.

Distributional shift: Safety apps trained on clean audio and video encounter adversarially degraded inputs in deployment. The adversary *is* the distribution shift.

Specification gaming: Social conditioning (L5) is a real-world instance of specification gaming: the adversary exploits the gap between what the system is specified to detect and what cultural context prevents the user from allowing it to detect.

10 Limitations

- Scenario library is not exhaustive. 14 scenarios cover urban/rural Indian context; generalisation requires expansion.
- Benchmark profiles are reference constructs, not validated against commercial products.
- ZIDR measurement is unvalidated at scale. Baseline thresholds require partner validation with safety app vendors.

- Degradation factors are benchmark priors for reproducible testing and are not direct empirical measurements from deployed commercial systems.
- Sociotechnical attack surface (L5) is qualitative; operationalising for quantitative ZIDR requires additional methodology.
- Tool specification only; full implementation is Phase 2.

11 Future Work

- ZIDR baseline measurement—controlled testing with 1–2 safety app vendors.
- Policy brief—targeted at CEN-CENELEC and India Bureau of Indian Standards.
- Expanded scenario library—coverage beyond Indian urban/rural context.
- L5 operationalisation—methodology for quantifying social conditioning in ZIDR.
- Academic submission—target ACM FAccT 2027.

12 Conclusion

Women’s safety AI systems face a threat that no existing benchmark tests and no existing governance framework governs: a physically proximate adversary who defeats passive detection without device access or technical knowledge. This paper provides four contributions to close that gap: a threat taxonomy grounding the attack surface; ZIDR as an operationalisable evaluation metric; a 14-scenario benchmark library; and empirical benchmark results establishing that governance compliance alone provides only marginal improvement over an undocumented baseline (+0.11), while adversarial hardening shows the only substantial gain (+0.23).

A system that achieves 95% benchmark accuracy and $ZIDR = 0.00$ is not a safe system. Making that distinction visible—and testable—is the policy contribution of this work.

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A Full Scenario Library (14 Scenarios)

All 14 scenarios. Urban (U-01 to U-09), Rural (R-01 to R-05). (*) = novel attack vector not documented in existing adversarial ML or GBV literature.

ID	Scenario	Min. Access	Attack Method	Layers Defeated	Governance Gap
U-01	Mumbai Local Train, Peak Hours	L0	Suppress + Corrupt	Vision, Audio, IMU	All: no transit sensor fusion standard
U-02	App-Based Cab, Night Route	L1	Intercept + Spoof	Sensor Fusion, Vision	All + India DPDP
U-03	Workplace, Authority Figure	L5	Suppress (social)	None (not initiated)	All: sociotechnical suppression ungoverned
U-04	Street Market, Bag Obstruction	L0	Suppress	Vision, Audio	All: no CV occlusion standard
U-05	Bus Stop, Exhaustion Attack (*)	L0	Exhaust	Response (human)	All: no alert exhaustion req.
U-06	Campus, Known Acquaintance	L1	Corrupt	Vision, NLP	All: NLP coercion bypass untested
U-07	Social Venue, Noise + Alcohol (*)	L0	Suppress + Corrupt	Audio, Vision	All: no high-noise certification
U-08	Public Toilet, Device Grab (*)	L2	Intercept	Comm, Response	All: no device interception standard
U-09	Transit Confinement, Moving Vehicle	L1	Suppress + Intercept	Vision, Audio, GPS, Comm	All: confined transit ungoverned
R-01	Agricultural Field, Isolated Worker	L0	Intercept + Suppress	Communication	EU AI Act, NIST, India DPDP
R-02	Village Pathway, Dusk	L0	Suppress	Vision, Response	All: no low-light rural CV standard
R-03	Isolated Road, Moving Vehicle	L0	Intercept + Corrupt	Sensor Fusion, Comm	Cross-jurisdictional: no rural connectivity req.
R-04	Village Common, Community Authority	L5	Suppress (social)	None (not initiated)	All + India IT Act
R-05	Agricultural Employer, Seasonal (*)	L5	Suppress (economic)	Response, Vision, Audio	All: economic coercion ungoverned

Calibration note: R-02/R-04 ZIDR = 0.92–0.95 (not 1.00)—perfect score is not academically defensible; minimal calibration uncertainty applied. U-03 ZIDR = 0.27–0.33: sociotechnical suppression is the dominant failure mode; passive-layer performance alone overstates system safety in authority contexts.

B Taxonomy Reference

See Table 1 (Section 3) for the full 4-layer $\times$ 5-method matrix with minimum access levels and scenario footnotes.

Scenario library (YAML):

`pretzelslab/ai-safety-research/womens_safety_adversarial/scenarios/`

Benchmark results (CSV):

`pretzelslab/ai-safety-research/womens_safety_adversarial/benchmark/results/`